

Reconstruction of Subsurface Temperature Field in the South China Sea From Satellite Observations Based on an Attention U-Net Model

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Abstract—In this study, an attention U-net network was proposed to reconstruct the subsurface temperature (ST) field with high temporal and spatial resolution in the South China Sea (SCS) from sea surface parameters observed by satellites. In addition to sea surface temperature, sea-level anomaly, and sea surface wind (SSW) field, the wind stress curl, which influences the 3-D structure of temperature through the induced Ekman pumping and transport, was also input into the model. The five-day average vertical temperature profiles with a spatial resolution of 0.5° from simple ocean data assimilation (SODA) reanalysis were used for training and evaluating the network. The results show that the attention U-net model performs quite well in ST reconstruction in the upper 100-m layers of the SCS. The additional input of wind stress curl helps to improve the model accuracy. The average root-mean-square error (RMSE)/bias of ST decreases from $1.08^\circ\text{C}/-0.21^\circ\text{C}$ to $1.01^\circ\text{C}/-0.05^\circ\text{C}$. In particular, the RMSE near the thermocline is reduced significantly by up to 10.9%. The estimation error of the attention U-net model is much smaller than that of some linear and tree models in the SCS, especially in shallow waters and regions with complex dynamic processes. The case study also shows that our model is capable of capturing the evolution of mesoscale processes in the SCS. The combination of satellite observations with a high-precision ST reconstruction model will help us comprehensively understand the fine structure and variation of temperature and circulation in the marginal seas and open oceans.

Index Terms—Attention mechanism, satellite observations, South China Sea (SCS), subsurface temperature (ST), U-net.

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I. INTRODUCTION

THE ocean temperature is a crucial parameter in the study of ocean dynamic processes at different temporal and spatial scales, such as the mesoscale eddies and internal waves. Its 3-D structure determines the characteristics of mixed layer and thermocline [1], [2], [3], [4], [5]. The commonly used field observation instruments for measuring temperature profiles in the ocean include shipborne conductivity, temperature, and depth (CTD), underwater mooring buoys, globally distributed Argo (Array for Real-time Geotrophic Oceanography) floats, and so on. However, it is difficult to obtain continuous and high-resolution subsurface temperature (ST) information with the limitation of *in situ* measurements. The changes in the interior ocean are still confusing due to the lack of subsurface observations.

Satellite remote sensing can provide valuable long-term data of many ocean surface variables with high spatial and temporal resolution, which meets the surface observation requirements of a variety of ocean phenomena. It is of great significance to associate the subsurface measurements with ocean surface variables for the study of ocean environment and dynamics. In the past few years, many methods have been developed to reconstruct 3-D ocean temperature fields based on satellite observations of surface parameters. Among them, the most widely used is the multiple linear regression (MLR) model [6], [7], [8], [9], [10], [11]. Carnes *et al.* [7] applied the empirical orthogonal function (EOF) to vertical temperature profiles in the Northwest Atlantic and Northwest Pacific Oceans and then established the polynomial relationship between ST and sea surface temperature (SST) and sea surface dynamic height from 1920 to 1991. Guinehut *et al.* [9] reconstructed a temperature anomaly field through the MLR method by merging monthly average Argo data profiles with a spatial resolution of 3.0° into sea surface data, including SST anomaly (SSTA) and sea surface dynamic height anomaly on 1.0° regular grid. The combination provided a good estimation of the temperature anomaly in the upper 200-m depth layer of the North Atlantic Ocean from 1995 to 1998 with the root-mean-square error (RMSE) varying from 0.1°C to more than 0.8°C in different geographical locations. Guinehut *et al.* [10] later carried out a similar study in the global ocean based on data collected between 1993 and 2009 and concluded that the RMSE between estimated temperature anomaly and *in situ*

observations changed from 0.5 °C at the surface to a maximum of 0.9 °C in the mixed layer. Jeong *et al.* [11] also used the MLR model to reconstruct the global 3-D ocean temperature from SSTA, sea-level anomaly (SLA), and sea surface wind (SSW) stress anomaly data with Argo measurements as label data. The RMSE of the estimated ST anomaly ranged from 0.25 °C near the surface to about 0.61 °C at 100-m depth.

In addition to the above statistical methods, the statistical-dynamic approach such as the surface quasi-geostrophic (SQG) method has also been used for 3-D temperature reconstruction, but in an indirect way. The SQG method was traditionally developed to retrieve subsurface density or 3-D geostrophic velocity in the ocean based on sea surface density, sea surface height (SSH), and historical buoyancy frequency profiles [12], [13]. By combining the SQG method and multivariate EOF analysis, Yan *et al.* [14] estimated the ocean density profiles from sea surface density and SSH and then obtained the vertical temperature structure based on the least-square multivariate EOF algorithm using SST, sea surface salinity (SSS), and the concurrent density profiles in the Northeast and Southeast Pacific Ocean.

More recently, with the rapid growth of satellite remote sensing data and ocean big data, the artificial intelligence (AI) approaches have shown a great potential for retrieving temperature profiles in the ocean [15], [16], [17], [18], [19], [20], [21]. Ali *et al.* [15] introduced an artificial neural network to reconstruct ST in the Arabian Sea from SST, SSH, SSW stress, radiation, and heat flux. The RMSE between the estimated temperature and the *in situ* observation at a fixed mooring was 0.58 °C. Wu *et al.* [16] obtained ST anomaly in the North Atlantic from SSTA, SSH anomaly, and SSS anomaly by applying a self-organizing map neural network. The monthly gridded Argo vertical temperature profile data with a spatial resolution of 1° were used for network training and labeling. The Argo dataset was also widely utilized as label data for estimating ST or ST anomaly from surface observations by means of various machine learning methods, such as support vector machine [17], random forest [18], clustering-neural network [19], multi-layer-perceptron network [21], extreme gradient boosting (XGBoost) [22], back-propagation neural network [23], convolutional neural network (CNN) [24], and convolutional long- and short-term memory [25]. Different combinations of surface variables have been input into these models, including SST/SSTA, SLA, SSS/SSS anomaly, SSW/SSW anomaly, absolute dynamic topography, and geostrophic current velocity, and the RMSE of the estimated ST varied from 0.04 °C to 1.68 °C in different regions of the global ocean.

Previous studies have shown that the interior temperature of the ocean can be estimated from surface observations within a relatively reasonable error range. Nevertheless, some issues remain in ST reconstruction:

- 1) Most studies used the monthly Argo temperature profile data with a relatively low spatial resolution of 1° as labels. In this way, either the fine structure or time evolution of ocean mesoscale/submesoscale processes with horizontal scales generally smaller than 100 km cannot be captured from the estimated sparse ST field, which

makes it difficult to fully understand the characteristics of these processes.

- 2) Compared with 3-D temperature reconstruction in the open oceans, few studies have paid attention to the marginal seas, such as the South China Sea (SCS), partly due to the lack of Argo data in some regions of these seas. In the SCS, the mesoscale and submesoscale processes occur frequently due to the monsoon, the complex topography, and intrusion of strong western boundary current Kuroshio [26], [27], [28], [29], [30], [31]. High-resolution 3-D observations are urgently needed to investigate their fine structures and track their generation and evolution. Therefore, it is very important and necessary to estimate ST with high spatial and temporal resolution in the SCS.

In this study, we aim to reconstruct a relatively high-resolution ST field in the upper 100 m of the SCS (0–23°N, 105–121°E) from satellite observations of surface variables by using the U-net deep learning model combined with the attention mechanism (hereinafter referred to as attention U-net model). SST, SLA, zonal/u component of SSW (SSWU) and meridional/v component of SSW (SSWV), which were commonly used in previous studies, were considered as model inputs. As is well known, SSW plays a dominant role in the movement of surface water through friction. The more important influence of the wind on the 3-D structure of temperature, salinity, and ocean circulation is through the SSW stress curl (SSWSC), which could induce the Ekman pumping effect and change the depth of the thermocline [32], [33]. To investigate its impact on the accuracy of temperature estimation, the SSWSC was also calculated and added to the model input. In the meantime, in order to increase the training samples and reconstruct the temperature field with higher spatial and temporal resolution, the five-day average vertical temperature profiles with a spatial resolution of 0.5° from simple ocean data assimilation (SODA) product were used for training and evaluating the network. The data have been proved to be reliable and widely applied to investigate different dynamic processes in the SCS [34], [35], [36], [37], [38], [39].

This article is organized as follows. The satellite observations of sea surface variables and ocean reanalysis data are described in Section II. Section III presents the configuration of the attention U-net ST reconstruction model in detail. The performance of the model in estimating the vertical temperature profiles of the SCS is shown in Section IV. The influence of wind stress curl input on the performance of the model and the possible reasons are also discussed. Section V shows a case study of applying the attention U-net model to ST reconstruction within ocean mesoscale eddies. The conclusions are given in Section VI.

II. DATA

A. Satellite Observations of Surface Variables

The SST data [Fig. 1(a)] used in this study are from the optimally interpolated SST (OISST) product produced by the National Oceanic Atmospheric Administration

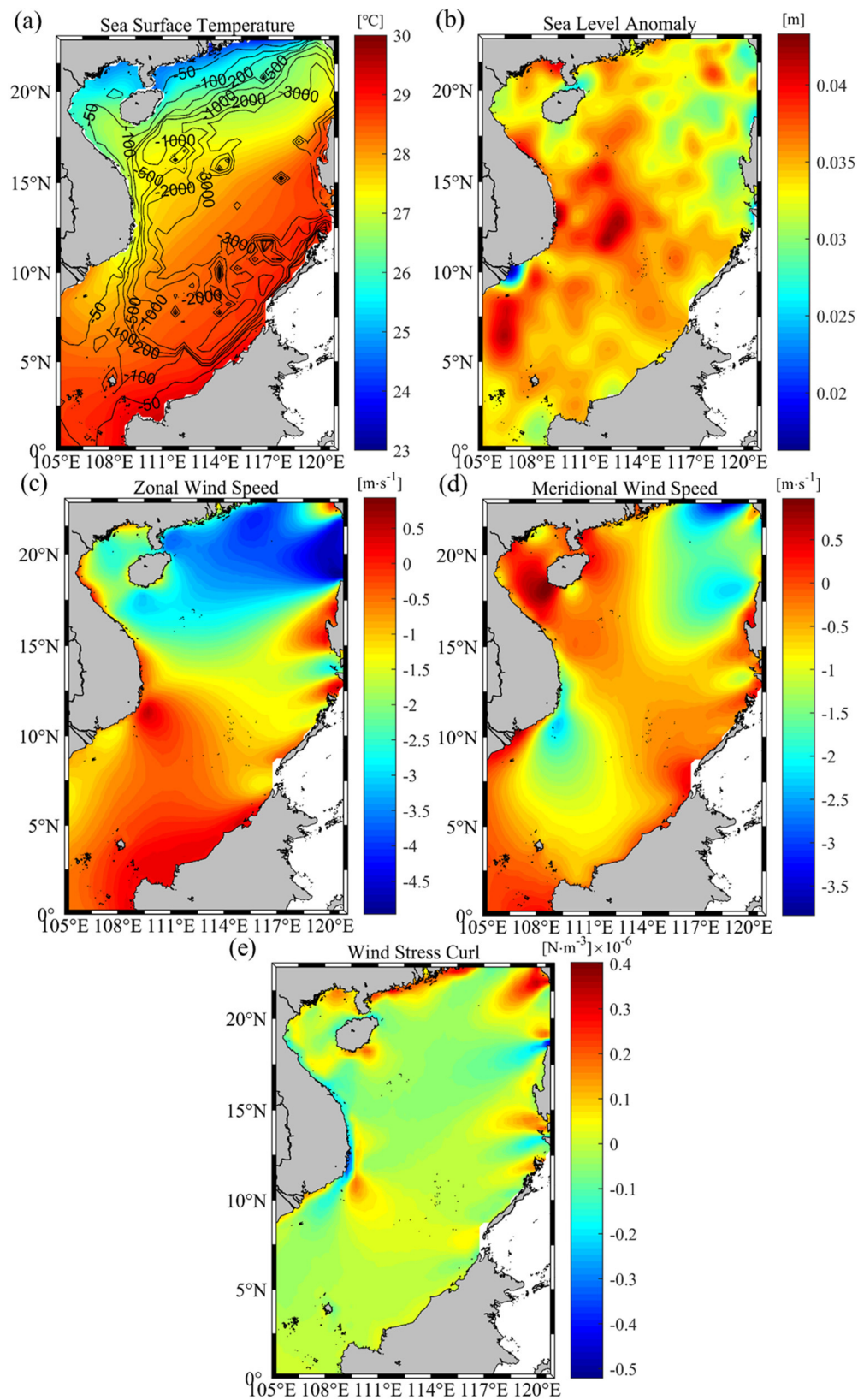


Fig. 1. Climatology of surface variables in the SCS from 1993 to 2017. (a) Sea surface temperature, (b) sea level anomaly, (c) zonal SSW speed, (d) meridional SSW speed, and (e) SSW stress curl. Contours of water depth (m) are overlaid in (a).

(NOAA) (<https://www.ncei.noaa.gov/products/optimum-interpolation-sst>). The OISST combines the observations collected from satellites, ships, buoys, and Argo floats

with the method of optimal interpolation, covering a wide range of time from September 1, 1981 to present. It is produced daily in near real time and has a spatial resolution

of 0.25° . The platform differences and sensor biases have been compensated with the reference of satellite and ship observations to buoys [40], [41].

The SLA product is based on all along-track altimeter missions, including Jason-3, Sentinel-3A, HY-2A, Saral/AltiKa, Cryosat-2, Jason-2, Jason-1, T/P, ENVISAT, GFO, and ERS1/2, and processed by the data processing system of Data Unification and Altimeter Combination System (DUACS). The multimission altimeter SLA data with a spatial resolution of 0.25° are available on the website of Copernicus Marine and Environmental Monitoring Service (CMEMS, <https://resources.marine.copernicus.eu>). Here, we used the daily and delayed time version of the product (id: SEALEVEL_GLO_PHY_L4_REP_OBSERVATIONS_008_047) [Fig. 1(b)].

The SSW data [zonal and meridional wind speeds, Fig. 1(c) and (d)] are from version 2 of Cross-Calibrated Multi-Platform (CCMP) wind product, which combines cross-calibrated satellite microwave winds and instrument observations using a variational analysis method to produce four maps daily of 0.25° gridded surface vector winds [42], [43]. The data can be downloaded from the Remote Sensing Systems (RSS, <https://www.remss.com/measurements/ccmp/>).

Based on the CCMP wind field, the wind stress curl [Fig. 1(e)] was calculated by

$$\begin{aligned} \text{Curl}\tau &= \frac{\partial \tau_y}{\partial x} - \frac{\partial \tau_x}{\partial y} \\ &= C_D \rho_a \left[U_{10} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) + \left(\frac{\partial U_{10}}{\partial x} v - \frac{\partial U_{10}}{\partial y} u \right) \right] \end{aligned} \quad (1)$$

where τ is SSW stress and τ_x and τ_y are its zonal and meridional components, respectively; C_D is the drag coefficient; ρ_a is the density of air; u and v are zonal and meridional wind speeds, respectively, and $U_{10} = (u^2 + v^2)^{1/2}$.

B. Vertical Temperature Profiles

The five-day average vertical temperature profiles with a spatial resolution of 0.5° are from version 3 of SODA (SODA3.12.2) reanalysis [44]. Compared with version 2, the model resolution, observation, forcing datasets, and the addition of active sea ice were all enhanced in version 3. The dataset produced and supplied by the University of Maryland consists of 50 vertical levels and covers a period from 1980 to 2017 (<https://dsrs.atmos.umd.edu/DATA/soda3.12.2/REGRIDED/ocean/>).

III. ATTENTION U-NET MODEL FOR ST RECONSTRUCTION

A. Attention U-Net Model

The U-net is one of the classic CNNs, which shows a strong ability of feature learning and representation [45], [46]. It is a modified fully convolutional network, which uses an image as an input and then outputs an image of the same size after the symmetric downsampling and upsampling processes. This yields a u-shaped architecture, which performs well on

the extraction of multiscale features of the image [47], [48]. The downsampling process includes the repeated application of two convolutions, each followed by an activation function and a max-pooling operation, which aims to capture the context. Meanwhile, the symmetric upsampling process is designed for precise localization and consists of upsampling operators, concatenations, and convolutions followed by the corresponding activation functions. The upsampling operators produce the features that can be combined with those from the downsampling process and increase the resolution of the output. The concatenations between the upsampled output and features from the contracting path help the network to localize. Besides, the U-net can be trained with very few training samples but yields the results with high accuracy [47], [49], [50]. This is of great significance for solving the small sample size problems.

Recently, the attention mechanism-based CNN, which combines the CNN with the attention model, has been presented. The attention mechanism focuses selectively on important features and can suppress unnecessary ones, which has been proven to be helpful to improve the accuracy of image classification or geophysical parameter estimation [51], [52], [53], [54], [55], [56]. In the attention model which is based on an encoder-decoder architecture, the attention weights of the inputs are produced for prioritizing the relevant input tokens while generating the next output token. This helps the model automatically learn to focus on the target and uses a small number of model parameters for efficient calculation. Though the U-net performs well on the multiscale feature extraction, it might repeatedly extract similar low-level features, which would lead to the redundant use of computational resources. The introduction of attention model can well balance this limitation by highlighting the salient features of the targets. Therefore, in this study, we combined the attention model with U-net to reconstruct the ST field in the SCS.

Among various attention models, the convolutional block attention module (CBAM), which emphasizes the attention in both channel and spatial dimensions [57], was selected for our attention U-net model. This would make each branch capable of learning “what” and “where” to attend in the channel and space, respectively. The structure and each submodule of CBAM are shown in Fig. 2(a). The CBAM includes the channel attention module (CAM) and spatial attention module (SAM). The CAM first uses the average- and max-pooling operations, which aims at aggregating spatial information and inferring finer channelwise attention, respectively, to obtain two different features. These features are merged by elementwise summation after the application of shared network. Then, an activation function is applied to produce the channel attention map. In the SAM, the elementwise multiplication between the output of CAM and the original input feature is used as an input. Two feature maps are obtained through average- and max-pooling operations along channels, which is effective in highlighting informative regions. The resulted features are concatenated and then put into a convolutional layer followed by an activation function to get the final attention map.



the feature downsampling, and the other is the application of CBAM, which amplifies the important features of the original sized object. The downsampled features are then double convoluted and input into the CBAM as well to capture the significant feature maps on the coarse object scale. The output is upsampled, concatenated with the previous output from the first CBAM, and double convoluted so that the number of feature maps is reduced by half. Finally, after a 1×1 convolution, a single feature map is obtained, which

represents the result produced by the network. In convolutional layers except for CBAM and output layer, the size of filters is 3×3 and the activation function is selected as “relu.” Meanwhile, the size of the filter is 7×7 and the activation function is “sigmoid” in the convolutional layer for CBAM.

B. Data Preprocessing

The input surface variables and vertical temperature profiles are required to have the same spatial and temporal resolution before training the network. Taking SODA data as a reference, we first calculated the five-day average of all surface variables and interpolated them to a spatial resolution of 0.5° . There are a total of 1824 samples from January 3, 1993 to December 19, 2017. Among them, 1753 samples before 2016 were used to train and validate the attention U-net model. The training and validation samples were randomly selected in the ratio of 9:1. The remaining 71 samples in 2017 were testing data.

The normalization of input features is necessary for the neural network model. It eliminates the difference between variables with different dimensions and magnitudes and makes them distributed in the same range of values after scaling. In this study, each surface variable was normalized as

$$X = \frac{x - \mu}{\sigma} \quad (2)$$

where x is the original surface variable of samples in training, validation, and testing datasets, μ and σ are the mean and standard deviation of x , respectively, and X is the normalized surface variable. All normalized variables would be clustered with the mean of 0 and the variance of 1.

C. Model Configuration

In this study, we focus on the reconstruction of the temperature structure in the upper 100 m of the SCS, which corresponds to the upper ten vertical layers of the SODA dataset. For each layer, a corresponding attention U-net model was established, so we obtained ten temperature estimation models altogether. As shown in Fig. 2(b), the estimation procedure is divided into three parts: input, training, and reconstruction. The input process is to prepare and preprocess the input surface variables. The inputs are then used to train the attention U-net by using SODA temperature profiles as labels. Finally, all surface variables will be input into the trained model to reconstruct the ST field. We selected “huber_loss” that can effectively reduce the penalty degree of deep learning model for outliers as the loss function and “adam” as the optimizer for the network.

IV. RESULTS AND DISCUSSION

A. Sensitivity Analysis of Model Input and Loss Function

SST, SLA, and SSW have been proved to be able to reconstruct ST with reasonable accuracy [17], [18], [19], [23]. What about the wind stress curl? In this section, a sensitivity analysis was carried out to investigate whether and how the SSWSC would affect the accuracy of ST estimation using the attention U-net model. Two comparative experiments were first designed, using surface variables with and without SSWSC as

model input, respectively. The RMSE, correlation coefficient (R), and mean bias between the retrieved temperature at each layer and SODA reanalysis on the testing dataset are shown in Table I and Fig. 3.

One can see that the errors vary with water depth in both experiments. The RMSE is the smallest at the top layer. It increases with the increase of water depth, reaches a maximum value at a certain depth (about 66 m), and then decreases slowly. The correlation coefficient decreases gradually from the top to deeper layers. The additional input of the SSWSC leads to a decline of RMSE at all layers, especially in waters deeper than 55 m, and the value is reduced by up to 10.9% at a water depth of 76.80 m. The mean bias is also much closer to 0 in deep waters and the average bias at all layers decreases significantly from -0.21°C to -0.05°C . The results indicate that the wind stress curl plays an important role in improving the estimation accuracy of the temperature in the upper layers of the SCS.

What results will we get if the SSWSC is the only model input? To answer this question, we carried out a third experiment. As shown in Table I or Fig. 3, the RMSE is much larger at layers with water depth less than 45 m, suggesting that the basic parameters, including SST, SLA, and SSW, are very important for accurate ST reconstruction in shallow waters. At lower layers (L06–L10), the error of the model is comparable to that with four or five surface variables as inputs. This further confirms the positive effect of the wind stress curl in the ST estimation in relatively deeper waters of the SCS. However, why is this effect more obvious in deeper waters? This will be discussed in Section IV-C.

In addition to the input variables, the loss function is another key parameter of the attention U-net model. The “huber” loss function used in this study can reduce the sensitivity of the model to the outliers, which would enhance its robustness for reconstructing ST in the SCS where invalid values are sometimes present in the label data. To illustrate the superiority of the “huber” loss function, sensitivity experiments were also conducted, in which “huber” and two other commonly used loss functions “mae” and “mse” were adopted. As shown in Table II, overall, the model performs much better with smaller RMSE and bias and slightly higher correlation coefficient when “huber” was used, especially in waters deeper than 55 m where there are more invalid ST values due to the complex topography of the SCS. In our model, the invalid ST value was specified as a number far away from the actual water temperature. Therefore, the use of “huber” loss function makes the model focus less on the outliers and thus improves the performance of the model.

B. Attention U-Net Model Performance Against Typical Models

Fig. 4 shows the density scatterplots of the temperature estimated from satellite observations of five surface parameters and SODA product. One can see that the attention U-net model performs well in capturing the vertical temperature structure in the upper layers of the SCS. The statistical errors on training and testing datasets are close to each other. The RMSEs

TABLE I

RMSE, MEAN BIAS, AND CORRELATION COEFFICIENT (R) BETWEEN ATTENTION U-NET ESTIMATED TEMPERATURE AND SODA REANALYSIS AT THE UPPER TEN VERTICAL LAYERS OF THE SCS. “SSWSC” REPRESENTS INPUT ONLY CONTAINING SSWSC. “4_var” AND “5_var” REPRESENT FOUR SURFACE VARIABLES (SST/SLA/SSWU/SSWV) AND FIVE SURFACE VARIABLES CONTAINING SSWSC (SST/SLA/SSWU/SSWV/SSWSC), RESPECTIVELY. THE RELATIVE DIFFERENCE OF RMSE IS $(RMSE5 - RMSE4)/RMSE4$ AND A NEGATIVE VALUE INDICATES AN IMPROVEMENT IN ACCURACY. POSITIVE AND NEGATIVE VALUES OF MEAN BIAS REPRESENT THE OVERESTIMATION AND UNDERESTIMATION OF ST, RESPECTIVELY

Layer No.	Depth/ m	RMSE/°C				Bias/°C			R		
		ssWSC	4_var	5_var	Relative difference	ssWSC	4_var	5_var	ssWSC	4_var	5_var
L01	5.03	0.83	0.39	0.38	-2.6%	0.21	0.03	0.03	0.92	0.98	0.98
L02	15.10	0.89	0.57	0.56	-1.8%	0.19	-0.04	-0.02	0.90	0.96	0.96
L03	25.22	0.97	0.77	0.76	-1.3%	0.12	-0.10	-0.11	0.89	0.94	0.94
L04	35.36	1.10	1.01	0.96	-5.0%	0.07	-0.24	-0.06	0.86	0.89	0.90
L05	45.58	1.30	1.19	1.14	-4.2%	0.03	-0.28	-0.24	0.80	0.85	0.86
L06	55.85	1.39	1.34	1.26	-6.0%	0.03	-0.33	-0.04	0.77	0.80	0.81
L07	66.26	1.39	1.42	1.28	-9.9%	-0.02	-0.26	0.13	0.73	0.74	0.78
L08	76.80	1.36	1.38	1.23	-10.9%	-0.06	-0.45	0.06	0.70	0.73	0.76
L09	87.58	1.29	1.26	1.18	-6.4%	-0.12	-0.34	-0.13	0.69	0.73	0.75
L10	98.62	1.23	1.17	1.10	-6.0%	-0.07	-0.21	-0.12	0.68	0.72	0.75
Average		1.18	1.08	1.01	-6.5%	0.05	-0.21	-0.05	0.93	0.95	0.95

TABLE II

RMSE, MEAN BIAS, AND CORRELATION COEFFICIENT (R) BETWEEN ATTENTION U-NET ESTIMATED TEMPERATURE AND SODA REANALYSIS AT THE UPPER TEN VERTICAL LAYERS OF THE SCS. “HUBER,” “MAE,” AND “MSE” LOSS FUNCTIONS WERE USED IN THE MODEL

Layer No.	Depth/ m	RMSE/°C			Bias/°C			R		
		huber	mae	mse	huber	mae	mse	huber	mae	mse
L01	5.03	0.38	0.38	0.38	0.03	0.12	0.01	0.98	0.98	0.98
L02	15.10	0.56	0.57	0.58	-0.02	-0.05	-0.03	0.96	0.96	0.96
L03	25.22	0.76	0.77	0.78	-0.11	-0.11	-0.09	0.94	0.94	0.93
L04	35.36	0.96	1.09	1.00	-0.06	-0.21	-0.03	0.90	0.87	0.89
L05	45.58	1.14	1.27	1.17	-0.24	-0.21	-0.02	0.86	0.82	0.85
L06	55.85	1.26	1.40	1.42	-0.04	-0.08	-0.04	0.81	0.77	0.77
L07	66.26	1.28	1.58	1.45	0.13	-0.51	-0.29	0.78	0.71	0.73
L08	76.80	1.23	1.51	1.47	0.06	-0.27	-0.47	0.76	0.68	0.71
L09	87.58	1.18	1.57	1.32	-0.13	-0.64	-0.35	0.75	0.66	0.71
L10	98.62	1.10	1.43	1.26	-0.12	-0.43	-0.17	0.75	0.63	0.68
Average		1.01	1.20	1.12	-0.05	-0.22	-0.13	0.95	0.93	0.94

(biases) are 1.02 °C (0.15 °C) and 1.01 °C (−0.05 °C), and the correlation coefficients are as high as 0.95, indicating that the model is quite stable. Most of the points in Fig. 4 are concentrated in the temperature range of 28 °C–30.5 °C, suggesting that the temperature estimation of warm waters is more accurate. The accuracy of our model is comparable to or a little bit higher than that of the other deep learning models in estimating the temperature within the upper 100-m layers of the open ocean [23], [24]. However, our study area is the marginal sea, where the accurate ST reconstruction may be more difficult due to its complex topography and dynamic processes.

Also, we compared the performance of the attention U-net model against the pure U-net model, the XGBoost model (one of the popular machine learning models), and the classical MLR model with the same input variables. As shown in Table III, the RMSEs of attention U-net estimated temperature are much smaller than those of XGBoost or MLR estimated temperature at all layers, and the average value is reduced by 33.6% and 36.9%. Meanwhile, the correlation between the ST field derived from the attention U-net model and the SODA product is higher, especially at the lower layers. When the attention module is added to the U-net model, although the correlation coefficient increases slightly, the average RMSE

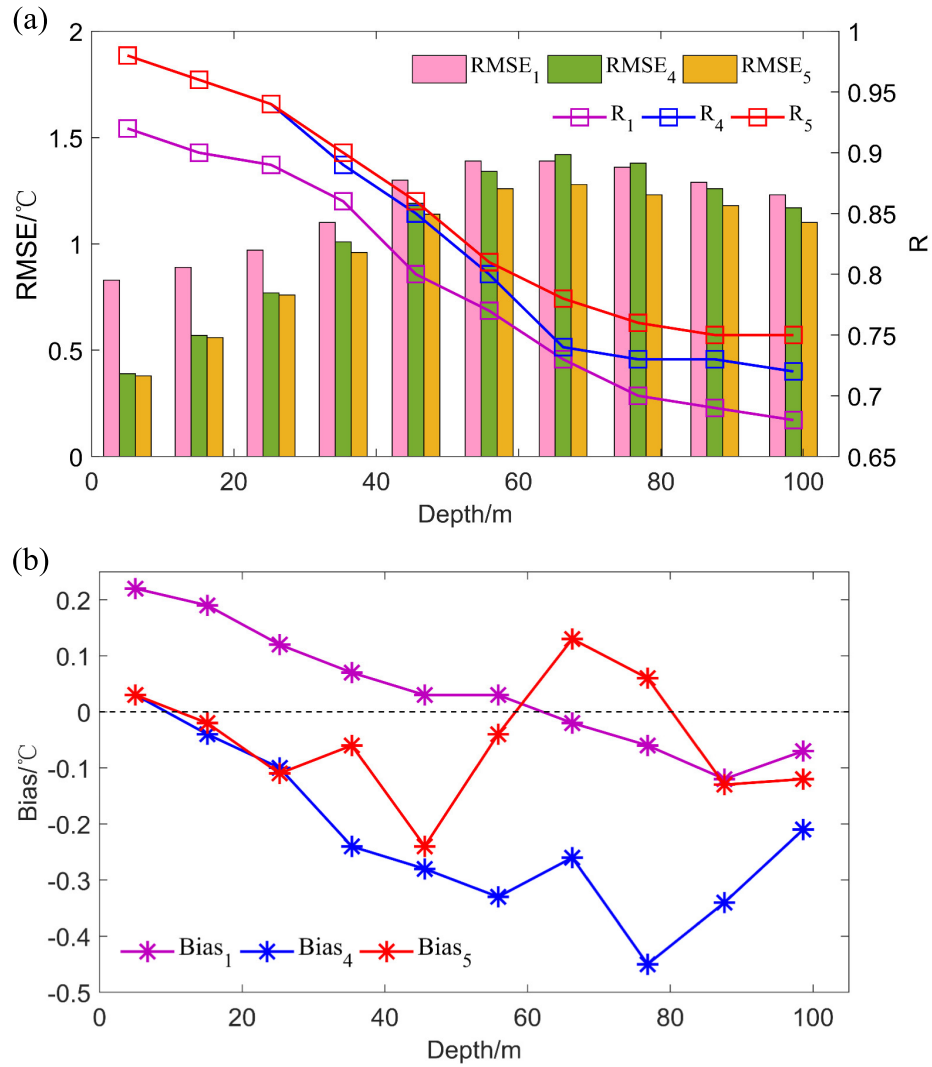


Fig. 3. Variation of (a) RMSE and R and (b) mean bias between attention U-net estimated temperature and SODA reanalysis as a function of water depth. The subscripts “1,” “4,” and “5” of RMSE, R , and bias represent the statistical errors of ST derived from only SSWSC, four surface variables without SSWSC, and five surface variables with SSWSC, respectively.

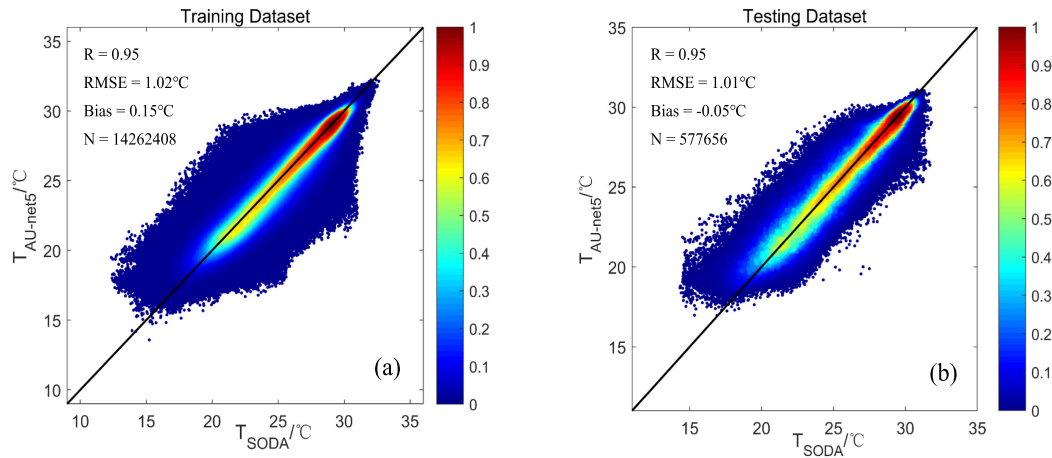


Fig. 4. Attention U-net model performance on (a) training and (b) testing datasets with five surface variables as inputs. The black line indicates the 1-to-1 line.

is reduced by 6.5% and the average bias decreases from -0.22°C to -0.05°C , indicating that the attention mechanism helps to generate ST with higher accuracy.

The horizontal distribution of RMSE estimated from different models is shown in Fig. 5. Compared with XGBoost and MLR models, the RMSE of ST derived from the U-net

TABLE III

RMSE, MEAN BIAS, AND CORRELATION COEFFICIENT (R) BETWEEN ATTENTION U-NET, U-NET, XGBOOST, OR MLR ESTIMATED TEMPERATURE AND SODA REANALYSIS AT THE UPPER TEN VERTICAL LAYERS OF THE SCS

Layer No.	Depth /m	RMSE/°C				Bias/°C				R			
		Attention U-net	U-net	XGBoost	MLR	Attention U-net	U-net	XGBoost	MLR	Attention U-net	U-net	XGBoost	MLR
L01	5.03	0.38	0.38	0.46	0.49	0.03	0.01	0.04	0.02	0.98	0.98	0.97	0.97
L02	15.10	0.56	0.59	0.75	0.84	-0.02	-0.05	-0.06	-0.08	0.96	0.96	0.93	0.91
L03	25.22	0.76	0.78	1.10	1.23	-0.11	-0.15	-0.17	-0.18	0.94	0.93	0.86	0.82
L04	35.36	0.96	1.02	1.42	1.55	-0.06	-0.24	-0.35	-0.36	0.90	0.89	0.77	0.72
L05	45.58	1.14	1.20	1.68	1.81	-0.24	-0.28	-0.51	-0.53	0.86	0.85	0.68	0.61
L06	55.85	1.26	1.33	1.87	1.98	-0.04	-0.30	-0.66	-0.70	0.81	0.80	0.58	0.52
L07	66.26	1.28	1.39	1.94	2.02	0.13	-0.28	-0.79	-0.85	0.78	0.75	0.49	0.44
L08	76.80	1.23	1.35	1.92	1.98	0.06	-0.36	-0.86	-0.91	0.76	0.73	0.43	0.38
L09	87.58	1.18	1.27	1.84	1.87	-0.13	-0.38	-0.82	-0.87	0.75	0.73	0.39	0.36
L10	98.62	1.10	1.14	1.72	1.72	-0.12	-0.22	-0.71	-0.76	0.75	0.74	0.37	0.37
Average		1.01	1.08	1.52	1.60	-0.05	-0.22	-0.46	-0.49	0.95	0.95	0.90	0.89

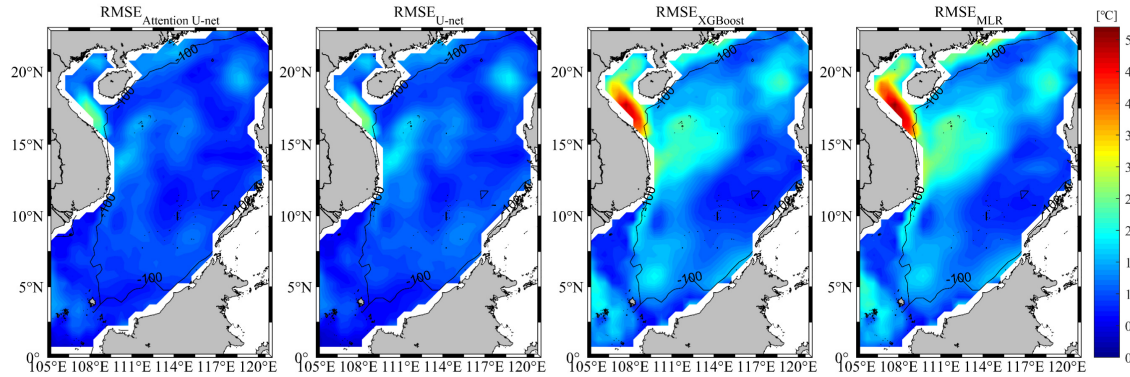


Fig. 5. Distribution of RMSE between attention U-net, U-net, XGBoost, MLR model estimated ST using five surface variables as inputs, and all SODA samples in the testing dataset in the SCS in 2017, superimposed with contour line of 200-m water depth.

or attention U-net model is much smaller in the whole SCS region. The reduction of the error is particularly significant in areas east of Vietnam where the coastal upwelling frequently occurs in summer [58] and the southwest of Luzon Strait where internal waves and mesoscale eddies are active [26], [27], [31], [59], [60]. The improvement in accuracy is also obvious in shallow waters with water depth of less than 200 m, such as coastal areas of China, Beibu Bay, and the southwest of Hainan Island, although the RMSE here is slightly larger than that in central parts of the SCS, which may be largely caused by the inaccurate observations of SLA by satellite altimeters in coastal regions due to the local departure of the radar signal from the known ocean response and the inaccuracy of the standard routinely applied corrections and tidal models in the product [61], [62]. The results imply that by considering the nonlinear relationship between surface variables and ST in both horizontal and vertical dimensions, the U-net type model helps to improve the estimation accuracy of temperature of the interior ocean with complex dynamic processes and topography. After introducing the attention mechanism into the U-net model, the ST estimation errors are further reduced,

especially in regions east of Vietnam and southwest of Luzon Strait.

Fig. 6 shows the spatial distribution of ocean temperature estimated by attention U-net, XGBoost, and MLR models at different depths (5.03, 35.36, 66.26, and 98.62 m) in the SCS averaged between December 12 and 16, 2017. Similar to the results in Table III, the differences between all model estimation results and SODA reanalysis are small in near-surface layers [Fig. 6(a) and (b)]. As the water depth increases [Fig. 6(c) and (d)], compared with XGBoost or MLR model, the spatial patterns of attention U-net derived temperature are more consistent with SODA product in most regions of the SCS.

To further demonstrate the good performance of our attention U-net model, we extended it to deeper layers. The estimation results of ST in the upper 450 m of the SCS are shown in Table IV. One can see that the RMSE decreases continuously with water depth and the value is as small as 0.31 °C at a water depth of about 450 m. The bias is also small, and the correlation coefficient slightly changes between 0.62 and 0.76. This illustrates that the model performs a little

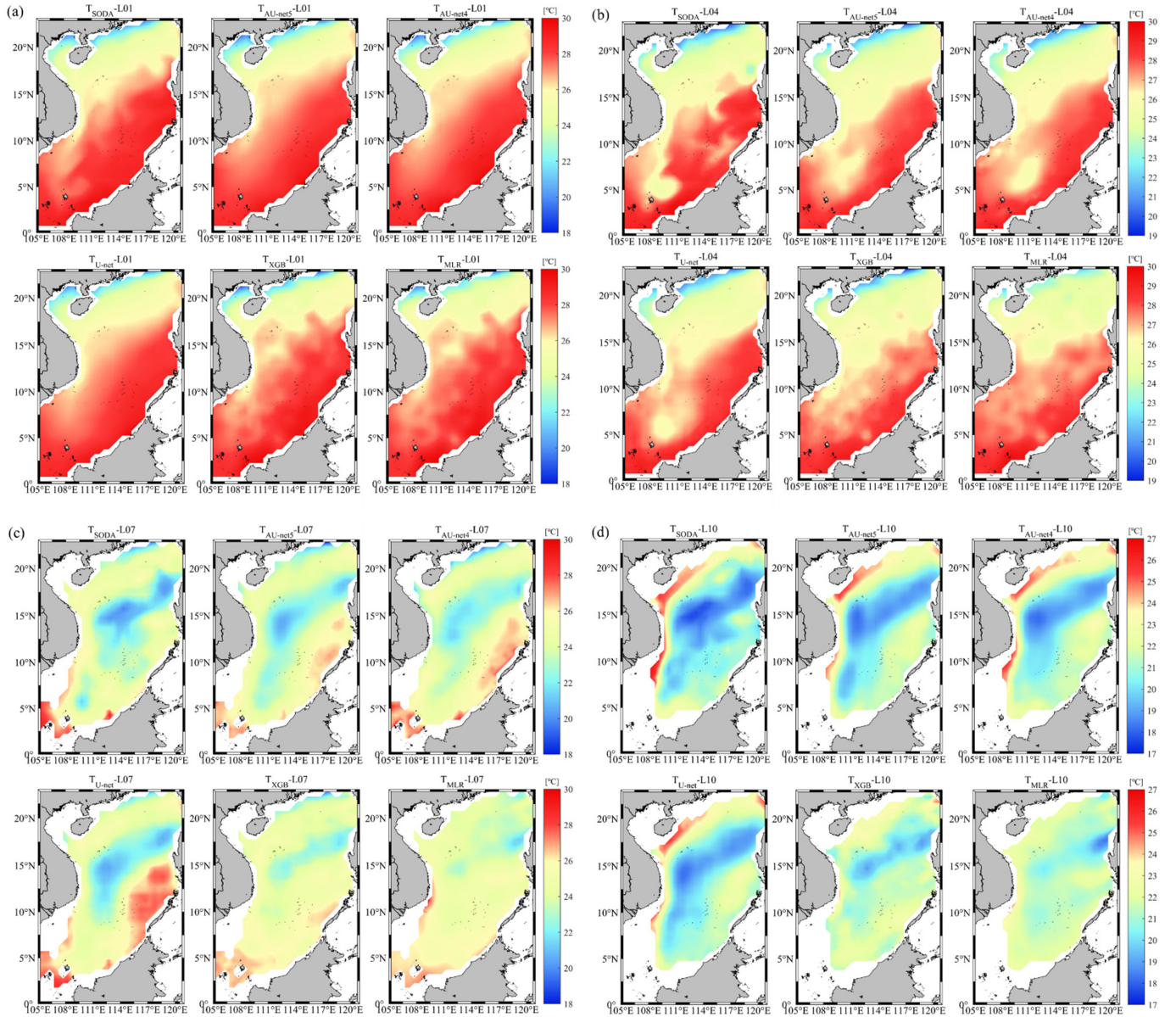


Fig. 6. Temperatures estimated by the attention U-net using four surface variables ($T_{AU-net4}$) as inputs and by the attention U-net ($T_{AU-net5}$), U-net (T_{U-net}), XGB (T_{XGB}), and MLR (T_{MLR}) using five surface variables as inputs compared to the SODA reanalysis (T_{SODA}) at layers (a) 1, (b) 4, (c) 7, and (d) 10 in the SCS averaged between December 12 and 16, 2017.

better in the ST reconstruction in deeper layers of the SCS, partly because the temperature structure there is more stable.

C. Influence of the Wind Stress Curl on Model Performance

In general, the attention U-net model performs quite well in the ST estimation in the SCS. Relatively high RMSE values occur at depths greater than 55 m without the wind stress curl input into the model (Table I and Fig. 3). Some researchers speculated that the larger error might be associated with the existence of thermocline where the temperature changes rapidly [15], [23]. To investigate whether the thermocline would affect ST estimation results, we calculated the thermocline depth and thickness of the study area in the testing

dataset. The combination of quasi-step function approximation method and gradient criterion method was used to extract the upper and lower boundary depths of the thermocline [63], [64]. The upper boundary depth and the difference between the two boundary depths are usually defined as the thermocline depth and thickness, respectively. Within the upper and lower boundaries, the depth with the maximum vertical temperature gradient is regarded as the maximum thermocline intensity depth.

Fig. 7 shows the spatial distribution of thermocline depth and thickness in the SCS in 2017. Fig. 8(a) presents the time series of regional averages of the thermocline depth, lower boundary depth, and maximum thermocline intensity depth. One can see that the thermocline is thicker in summer and

TABLE IV
THE RMSE, MEAN BIAS AND CORRELATION COEFFICIENT (R) BETWEEN ATTENTION U-NET ESTIMATED TEMPERATURE AND SODA REANALYSIS AT VERTICAL LAYERS SHALLOWER THAN 450 m OF THE SCS

Layer No.	Depth /m	RMSE /°C	Bias /°C	R	Layer No.	Depth /m	RMSE /°C	Bias /°C	R
L11	110.10	1.06	-0.17	0.75	L18	224.77	0.72	0.03	0.72
L12	122.11	1.00	-0.12	0.76	L19	253.07	0.66	0.01	0.71
L13	134.91	0.94	-0.09	0.76	L20	287.55	0.57	0.06	0.70
L14	148.75	0.91	-0.03	0.75	L21	330.01	0.51	0.01	0.66
L15	164.05	0.87	0.03	0.76	L22	382.37	0.39	0.03	0.62
L16	181.31	0.83	0.06	0.74	L23	446.73	0.31	0.07	0.64
L17	201.26	0.77	0.06	0.72					

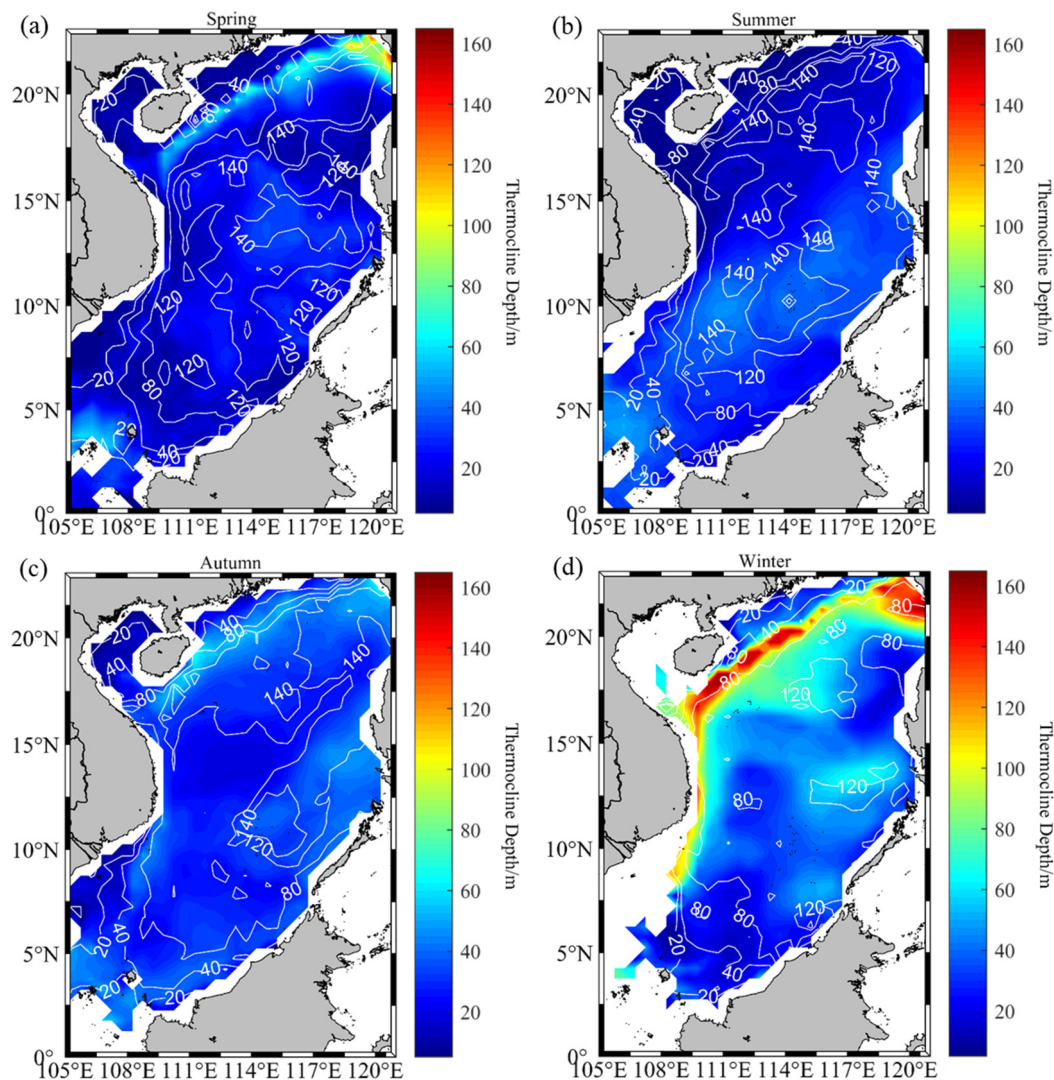


Fig. 7. Distribution of thermocline depth in (a) spring, (b) summer, (c) autumn, and (d) winter in 2017, superimposed with white contours of thermocline thickness (m).

thinner in winter. The average maximum thermocline intensity depth ranges from about 45 m in May to 85 m in February, which is consistent with the depth where the relatively larger RMSE of ST arises. This indicates that the thermocline does

have a significant influence on ST reconstruction. As mentioned earlier, the additional input of wind stress curl in the attention U-net model significantly improves the accuracy of the estimated temperature in the upper layers deeper than 55 m

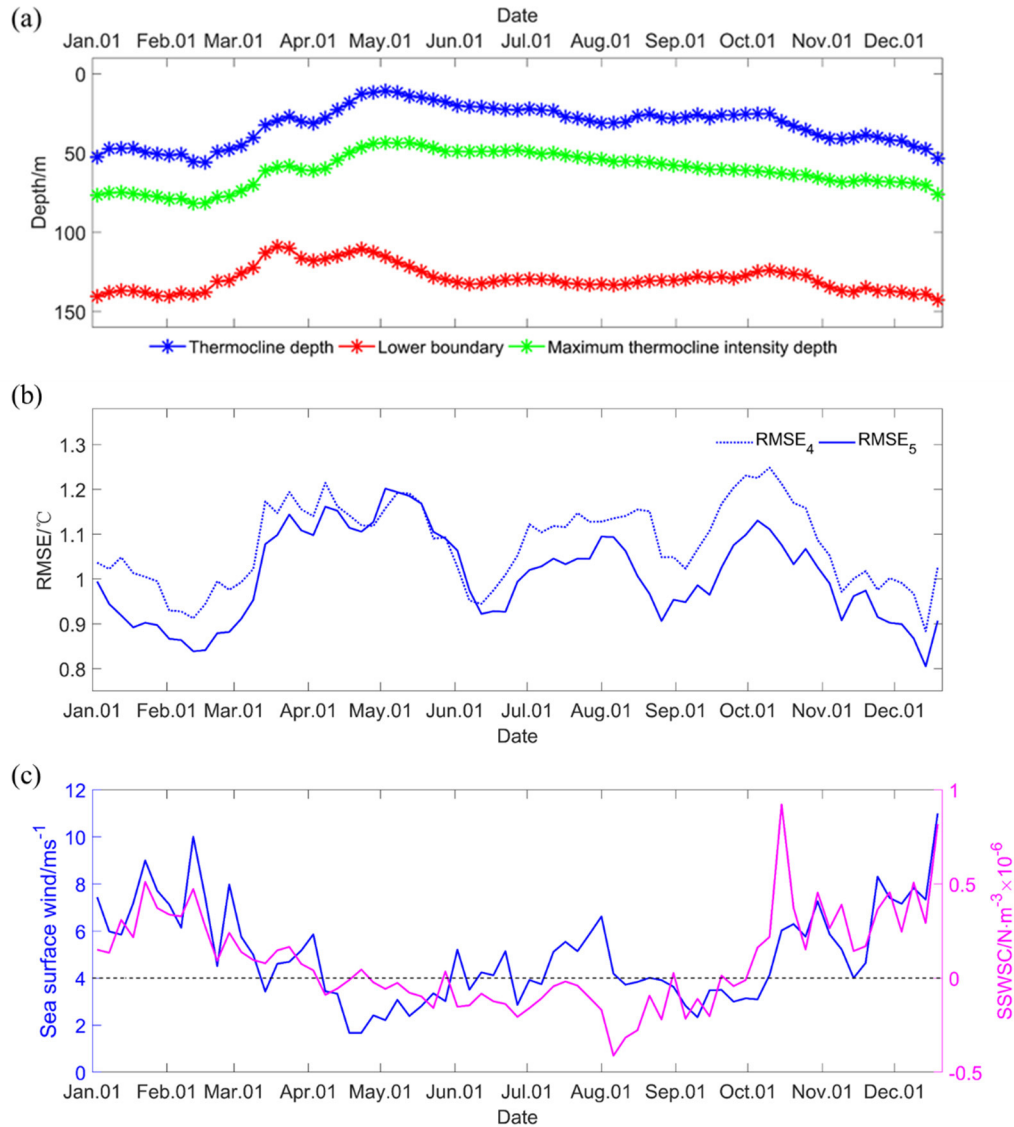


Fig. 8. Variation of (a) upper boundary depth (i.e., thermocline depth) and lower boundary depth of the thermocline and the maximum thermocline intensity depth, (b) RMSE between attention U-net estimated ST and SODA product, and (c) SSW and wind stress curl over time in the SCS in 2017. The black dashed line in (c) is the zero line of wind stress curl.

(Table I and Fig. 3). Therefore, it is possible that the wind stress curl deepens or shallows the thermocline through Ekman pumping and transport, which changes the thermal structure of waters near the maximum thermocline intensity depth and thus affects the results of ST reconstruction.

The temporal change of RMSE for all samples in the testing dataset is shown in Fig. 8(b). In general, the RMSE is smaller in winter when the ocean mixing is strong due to the relatively strong northeast monsoon. The thermocline is usually deeper than 50 m during this period [Figs. 7(d) and 8(a)]. The RMSE is a little larger from mid-March to the end of October when the thermocline becomes shallower and thicker. This brings more difficulties to accurately estimate ST due to the influence of thermocline on the thermal structure of the upper layers. Therefore, in the reconstruction of ST in the SCS where the thermocline demonstrates obvious temporal and spatial variations, it is suggested to introduce new input variables that may affect the thermocline. Several studies have shown

that the wind stress curl would strongly influence the upper ocean temperature structure by the induced Ekman pumping effect [32], [33], [65], [66], [67]. In our study, the accuracy of ST estimation is apparently improved when the impact of the wind stress curl is considered, especially from June to October, when the thermocline is shallower and thicker. In this case, the influence of the wind stress curl is easier to reach the thermocline depth. However, the decrease of RMSE is not obvious in April and May. As shown in Fig. 8(c), this may be because the SSW is the smallest during this period and, therefore, the wind stress curl is very weak and close to zero, which has little effect on the temperature profiles.

D. Estimated High-Resolution ST Against Low-Resolution Data

It is difficult to capture the fine structures of ocean mesoscale/submesoscale processes by using only low-resolution observations, such as the Argo float data. To fully

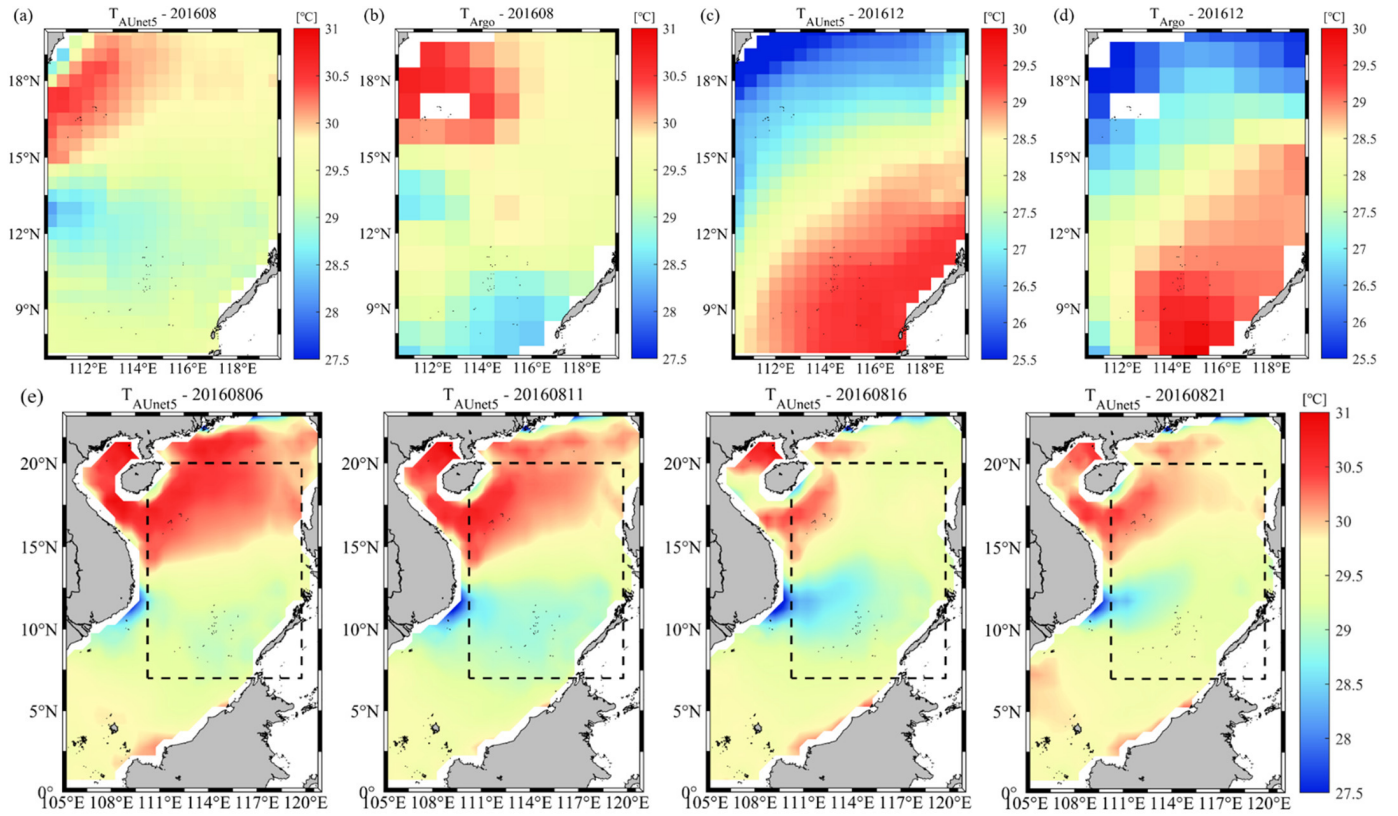


Fig. 9. High-resolution attention U-net estimated temperature at the depth of 5.03 m in the subregion of the SCS in (a) August and (c) December 2016, (b) and (d) low-resolution Argo observed temperature at the depth of 5 m in the same region and months, and (e) estimated temperature at the depth of 5.03 m in the SCS from August 6, 2017 to August 21, 2017. The dashed box in (e) marks the subregion of the SCS in (a)–(d).

understand the thermal and dynamic characteristics of these processes and their evolution, observations with higher spatial and temporal resolution are necessary. Here, the attention U-net estimated high-spatial ST fields were compared with the monthly Barnes objective analysis Argo (BOA-Argo) data on the 1° regular grid provided by the China Argo Real-time Data Center (<http://www.argo.org.cn>). The data consist of temperature and salinity measurements with 49 vertical levels from the surface to 1950 m depth and cover a time range from 2004 to 2017 [68]. In the SCS, the gridded BOA-Argo data are only available in the eastern region ($7\text{--}20^\circ\text{N}$, $110\text{--}120^\circ\text{E}$) from 2004 to 2016. Thus, we only discuss the difference between ST with high and low resolution in this subregion. The ST results reconstructed from satellite observations of surface variables were averaged by month for comparison with monthly Argo data.

Fig. 9(a)–(d) presents the estimated high-resolution ST at the depth of 5.03 m in the subregion of the SCS in August and December 2016, and Argo observed low-resolution ST at 5-m depth. The ST fields from the two datasets show similar spatial patterns, but there is a lack of observation in some regions, especially coastal areas, in Argo data. In addition, the reconstructed ST maps contain more information. For example, the detailed structure of the cold waters centered at (12°N , 110°E) is obvious in Fig. 9(a) but not so clear in Fig. 9(b). As shown in the five-day averaged reconstructed temperature maps in Fig. 9(e), the cold waters originated

along the Vietnamese coast and stretched northeastward into the central SCS from August 6, 2017 to August 21, 2017, indicating the existence of a coastal upwelling that usually occurs off eastern Vietnam in summertime [67], [69], [70]. The fine thermal structure and evolution of the coastal upwelling can be more accurately reproduced through higher spatial and temporal resolution data, which helps us better understand the dynamic mechanism of the process.

E. Robustness of the Attention U-Net Model

In this section, we try to extend our attention U-net model to reconstruct ST in the SCS with spatial and temporal resolution higher than that of SODA reanalysis by using daily and 0.25° gridded temperature profiles as label data, which are from the GLORYS2V4 reanalysis product provided by CMEMS (product id: GLOBAL_REANALYSIS_PHY_001_031). Data from 2010 to 2016 were selected as label data, and those in 2017 were used to evaluate the model accuracy. The model inputs are satellite observations of five surface variables with the same resolution. The RMSE, mean bias, and correlation coefficient between the reconstructed ST and GLORYS2V4 reanalysis product at the upper 100 m layers with water depth like that of the SODA product are shown in Table V. Compared with Table I, it can be seen that no matter which reanalysis dataset was used to label and train the network, our attention U-net model can reproduce the ST field

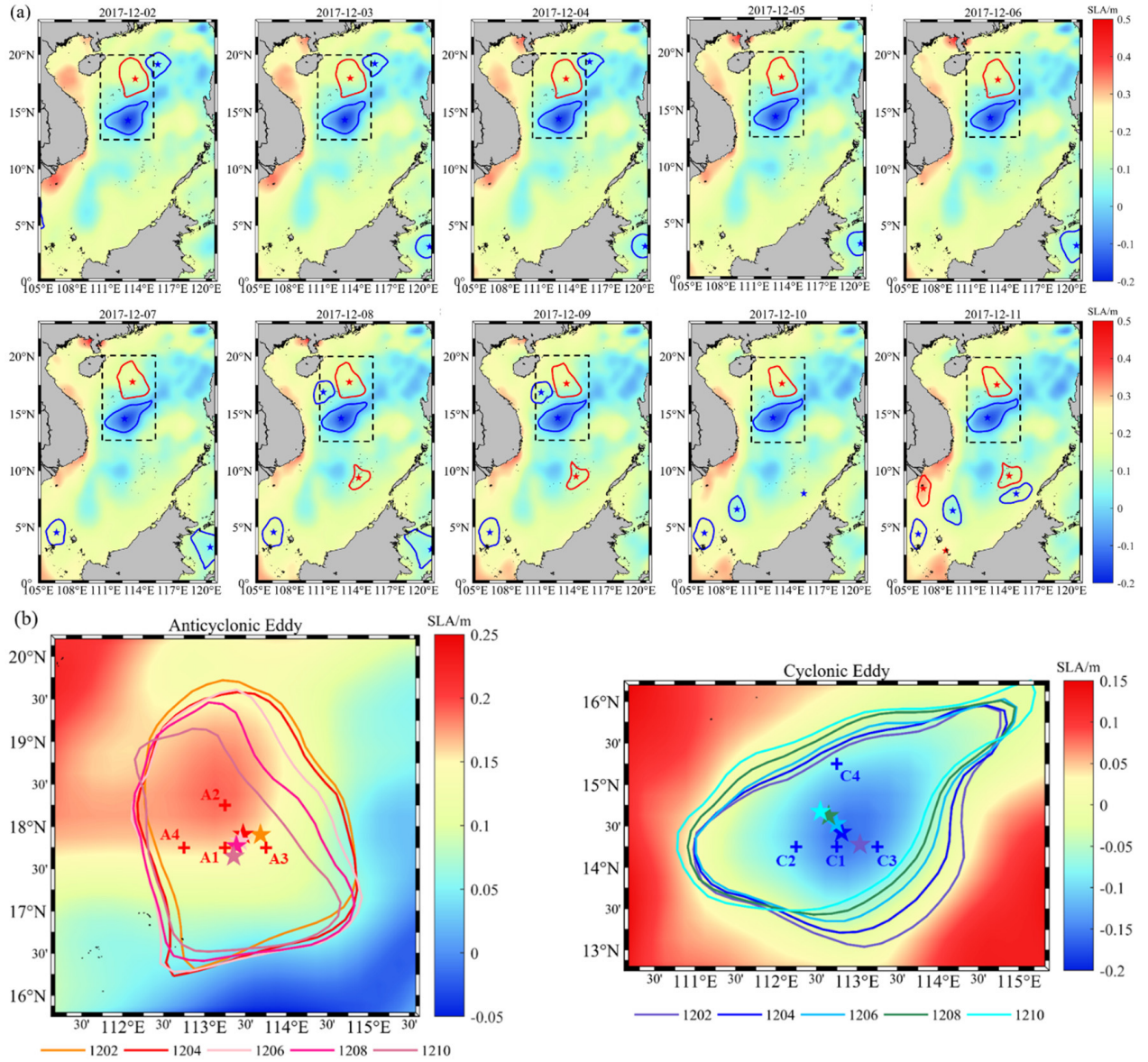


Fig. 10. (a) Daily SLA map of the SCS from July 2, 2017 to 11, 2017. The dotted black box marks the eddies investigated in this study. The blue star and line denote the center and contour of the cyclonic eddy, respectively, and the red star and line denote that of the anticyclonic eddy. (b) Change of the anticyclonic and cyclonic eddy boundaries (lines) and center locations (stars) within ten days, superimposed on the ten-day average SLA field. A1–A4 and C1–C4 marked with the plus sign represent four randomly selected positions in the anticyclonic and cyclonic eddies, respectively.

fairly well, indicating the model has strong robustness. The average RMSE of the estimated higher resolution temperature is a little smaller, implying that the model may learn more features from the finer structure of ST in the training process.

V. APPLICATION OF THE ATTENTION U-NET MODEL IN ST RECONSTRUCTION WITHIN OCEAN MESOSCALE EDDIES

The mesoscale eddies occur frequently in the SCS, especially near the southwest of Luzon Strait, which could significantly affect the vertical structure of ocean temperature [26], [59]. To evaluate the performance of the attention U-net model in ST reconstruction within the mesoscale eddies, a cyclonic eddy and an anticyclonic eddy were selected at the same time for analysis. The information of mesoscale

eddies was extracted from the altimetric Mesoscale Eddy Trajectories Atlas (META3.1exp DT allsat version), which was produced by SSALTO/DUACS and distributed by AVISO+ (<https://aviso.altimetry.fr>) with support from CNES, in collaboration with IMEDEA [71], [72]. The outermost closed SLA or absolute dynamic topography contour where the mean geostrophic currents are maximum is often regarded as the eddy boundary in order to eliminate the influence of strong deformation on the boundary induced by the surrounding currents on eddy characteristics [73]. The lifespan of both cyclonic and anticyclonic eddies should be longer than five days so that the signals of mesoscale eddies would not disappear after a five-day average of surface parameters.

Fig. 10(a) shows the variation of SLA in the SCS from December 2, 2017 to December 11, 2017. The evolution of

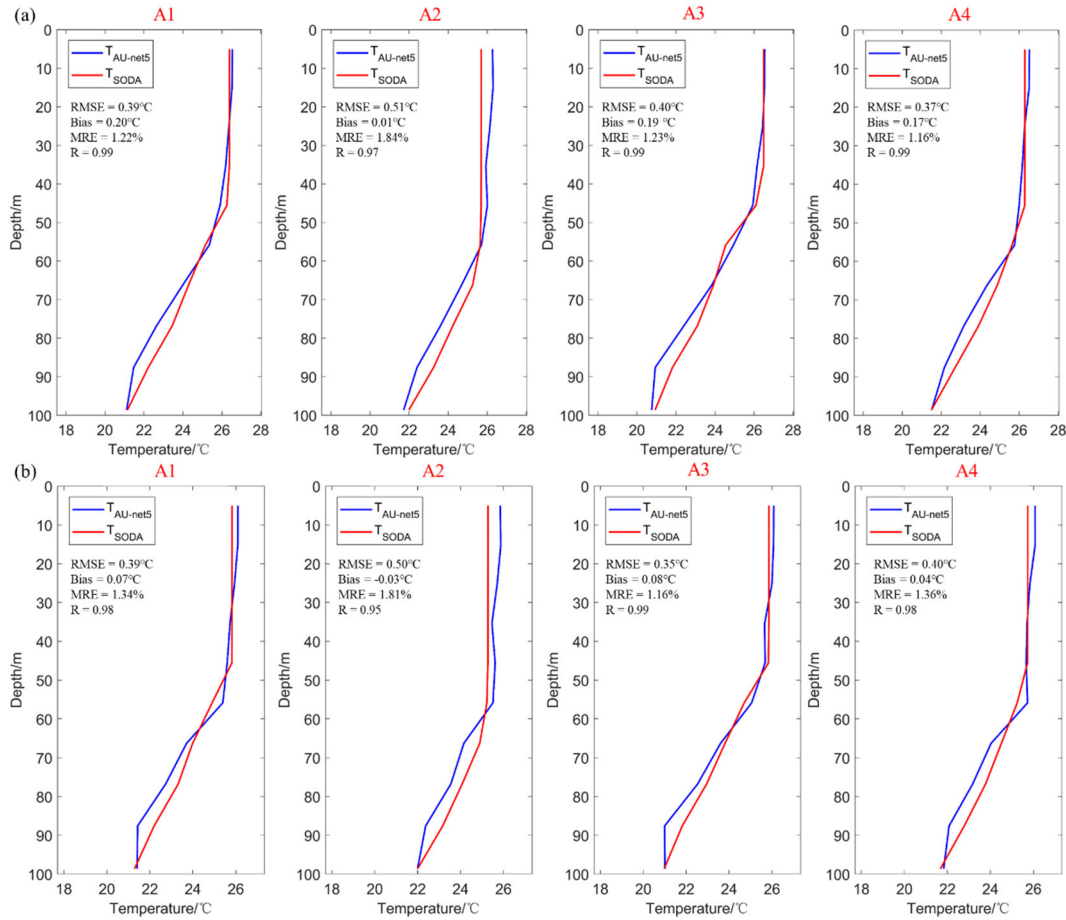


Fig. 11. Attention U-net model estimated ST compared to SODA temperature profiles within the anticyclonic mesoscale eddy averaged (a) between December 2 and 6, 2017 and (b) between December 7 and 11, 2017. Mean relative error (MRE) is the average of the relative error.

TABLE V
RMSE, MEAN BIAS, AND CORRELATION COEFFICIENT (R) BETWEEN ATTENTION U-NET ESTIMATED TEMPERATURE AND GLORYS2V4 REANALYSIS AT THE UPPER TEN VERTICAL LAYERS OF THE SCS

Layer No.	Depth/m	RMSE/°C	Bias/°C	R
L05	5.14	0.44	-0.03	0.97
L10	13.99	0.52	-0.06	0.96
L14	26.56	0.75	-0.12	0.93
L16	35.74	1.00	-0.26	0.87
L18	47.21	1.14	-0.25	0.83
L19	53.85	1.16	-0.18	0.80
L21	69.02	1.16	-0.09	0.75
L22	77.61	1.01	0.02	0.80
L23	86.93	0.96	0.05	0.80
L24	97.04	0.94	0.05	0.78
Average		0.92	-0.09	0.96

various mesoscale eddies with radius larger than 100 km is present on SLA maps. An anticyclonic eddy and a cyclonic eddy in the dotted box are always active in these ten days, which were selected to analyze the estimation accuracy of ST within the mesoscale eddies. Changes of their boundaries

and center locations are shown in Fig. 10(b). One can see that the anticyclonic eddy propagates southwestward, during which the eddy radius becomes smaller, while the cyclonic eddy propagates northwestward and the eddy-induced SLA signal is gradually weakened. Their shapes have changed during the propagation. Here, four positions always inside the anticyclonic or cyclonic eddy were randomly selected for model performance analysis.

As shown in Fig. 11, the model estimated ST within the anticyclonic eddy and its evolution agree well with SODA temperature profiles in the upper 100 m of the SCS. The RMSE ranges from 0.35 °C to 0.51 °C, the mean bias is less than 0.20 °C, and the correlation coefficient is greater than 0.95 (up to 0.99). The differences between the model estimation and SODA dataset change only slightly in ten days, demonstrating the good performance of attention U-net model in tracking the variation of ST within the anticyclonic eddy during its propagation. As for the temperature profiles within the cyclonic mesoscale eddy (Fig. 12), the estimation is also in good agreement with the SODA dataset, even though the errors between them are a little bit larger than those inside the anticyclonic eddy. The RMSEs of the upper layer ST at all grid points always inside the anticyclonic eddy and cyclonic eddy are 0.47 °C and 0.77 °C, respectively. Meanwhile, similar

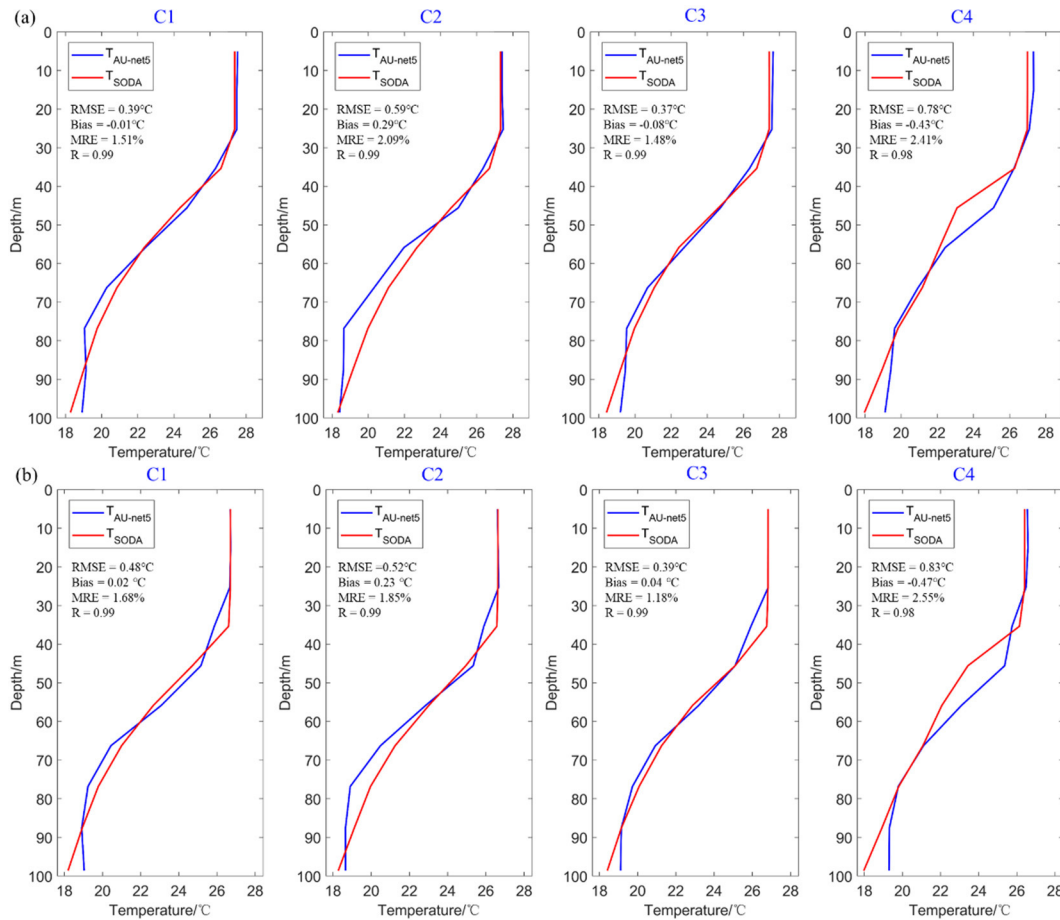


Fig. 12. Attention U-net model estimated ST compared to SODA temperature profiles within the cyclonic mesoscale eddy averaged (a) between December 2 and 6, 2017 and (b) between December 7 and 11, 2017. Mean relative error (MRE) is the average of the relative error.

results in Fig. 12(a) and (b) show that the attention U-net model can also well capture the change of ST within the propagating cyclonic eddy.

VI. CONCLUSION

This study aims to provide a useful technique for the study of 3-D temperature structure of the marginal sea by reconstructing the ST field with relatively high spatial and temporal resolution from satellite observations of sea surface parameters using the AI approach. Here, we introduced an attention U-net model that combines the attention mechanism with U-net to estimate ST in the upper 100-m water layers of the SCS using SST, SLA, SSWU, SSWV, and SSWSC. The five-day average temperature profiles with a spatial resolution of 0.5° from the SODA product were used for training the network and evaluating the accuracy of the estimated results.

The input sea surface variables are crucial for ST reconstruction from the attention U-net model. Both the RMSE and bias between the estimated temperature and SODA reanalysis at all vertical layers are reduced to a certain extent when the wind stress curl was added to the model input. The average RMSE decreases from 1.08°C to 1.01°C and the mean bias decreases from -0.21°C to -0.05°C , with the overall correlation coefficient remaining at 0.95. In particular, the RMSE of the

temperature near the thermocline is reduced significantly by up to 10.9% at a water depth of 76.8 m (L08). This indicates that the wind stress curl plays an important role in improving the reconstruction accuracy of ST by influencing 3-D structure of temperature through the induced Ekman pumping and transport.

The statistical analysis of the derived temperature on both training and testing datasets shows that our model is quite stable in the ST estimation. The average RMSE of attention U-net derived ST is 6.5%, 33.6%, and 36.9% smaller than that estimated from the pure U-net model, the XGBoost machine learning model, and the classical MLR model with the same inputs, respectively. The decrease of estimation errors is present particularly in shallow waters and regions with complex dynamic processes such as coastal upwelling, internal waves, and mesoscale eddies, suggesting that the ability of the attention U-net model to consider 3-D and nonlinear relationship between surface variables and ST is helpful to improve the accuracy of ST estimation in these areas. Compared with the open ocean, the dynamic processes and topography are usually more complicated in the SCS, which may bring more difficulty in accurate ST reconstruction. However, the average accuracy of our model is still comparable to or slightly higher than that of the other machine learning models that were used

to estimate the temperature in the upper 100-m layers of the open ocean [23], [24].

Compared with low-resolution temperature profile data, the high-resolution reconstructed ST from the attention U-net model can reproduce the fine 3-D thermal structure and evolution of the mesoscale/submesoscale dynamic processes in the SCS such as the coastal upwelling off eastern Vietnam and widely distributed mesoscale eddies.

In general, the attention U-net model is effective and shows strong robustness in ST reconstruction in the SCS with relatively high spatial and temporal resolution. This will help us understand the characteristic and evolution of the internal dynamic processes in the marginal sea more comprehensively. With the upcoming launch of new-generation satellites, such as the Surface Water and Ocean Topography (SWOT) mission, we will be able to observe the sea surface variables with an unprecedented resolution [74]. The performance of the ST estimation model based on satellite observations in the marginal seas is expected to be further improved.

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